EXPERT OPINION ON DRUG METABOLISM & TOXICOLOGY, 2018
VOL. 14, NO. 1, 25–41
https://doi.org/10.1080/17425255.2018.1416097

REVIEW

Side effects associated with current and prospective antimigraine pharmacotherapies

Abimael González-Hernández [a], Bruno A. Marichal-Cancino [b], Antoinette MaassenVanDenBrink [c] and Carlos M. Villalón [d]

[a]Instituto de Neurobiología, Universidad Nacional Autónoma de México, Querétaro, México; [b]Departamento de Fisiología y Farmacología, Universidad Autónoma de Aguascalientes, Ciudad Universitaria, Aguascalientes, México; [c]Division of Vascular Medicine and Pharmacology, Department of Internal Medicine, Erasmus University Medical Center, Rotterdam, The Netherlands; [d]Departamento de Farmacobiología, Cinvestav-Coapa, Ciudad de México, México

ABSTRACT

Introduction: Migraine is a neurovascular disorder. Current acute specific antimigraine pharmacotherapies target trigeminovascular $5\text{-HT}_{1B/1D}$, 5-HT_{1F} and CGRP receptors but, unfortunately, they induce some cardiovascular and central side effects that lead to poor treatment adherence/compliance. Therefore, new antimigraine drugs are being explored.

Areas covered: This review considers the adverse (or potential) side effects produced by current and prospective antimigraine drugs, including medication overuse headache (MOH) produced by ergots and triptans, the side effects observed in clinical trials for the new gepants and CGRP antibodies, and a section discussing the potential effects resulting from disruption of the cardiovascular CGRPergic neurotransmission.

Expert opinion: The last decades have witnessed remarkable developments in antimigraine therapy, which includes acute (*e.g.* triptans) and prophylactic (*e.g.* β-adrenoceptor blockers) antimigraine drugs. Indeed, the triptans represent a considerable advance, but their side effects (including nausea, dizziness and coronary vasoconstriction) preclude some patients from using triptans. This has led to the development of the ditans (5-HT_{1F} receptor agonists), the gepants (CGRP receptor antagonists) and the monoclonal antibodies against CGRP or its receptor. The latter drugs represent a new hope in the antimigraine armamentarium, but as CGRP plays a role in cardiovascular homeostasis, the potential for adverse cardiovascular side effects remains latent.

ARTICLE HISTORY
Received 24 October 2017
Accepted 7 December 2017

KEYWORDS
Adverse side effects; antimigraine drugs; CGRP; headache; lasmiditan; migraine; serotonin (5-hydroxytryptamine; 5-HT); triptans

1. Introduction

Migraine is a neurovascular disorder that consists of attacks of a severe and throbbing unilateral headache associated with nausea and vomiting, photophobia and phonophobia as well as neurological symptoms [1]. According to the third edition of the International Classification of Headache Disorders [ICHD, 2013], migraine has two major subtypes, namely: (i) migraine without aura (characterized by headache with specific features and associated symptoms) and (ii) migraine with aura (in which the headache is preceded or accompanied by transient focal neurological symptoms).

From an epidemiological perspective, migraine has been ranked as the third most prevalent disorder and the seventh highest specific cause of disability worldwide [2], and it represents an enormous socioeconomic burden to the individual as well as to the society [3].

1.1. Pathophysiology of migraine

Migraine is considered a neurovascular disorder that comprises activation of the trigeminovascular system, with the primary dysfunction located in brainstem centers regulating vascular tone and pain sensation. However, theories suggesting that the initiation of a migraine attack involves a primary event at supraspinal levels cannot be categorically excluded [1,4,5]. Indeed, as reviewed by Goadsby et al. [6], thalamic and/or cortical dysfunction of pain processing arriving from the trigeminovascular region could play a role. Moreover, dysfunction of hypothalamic (posterior, paraventricular, and A11 areas) modulation of trigeminovascular pain pathways seems to be relevant. Certainly, the fact that premonitory symptoms may appear prior to headache suggests that diencephalic structures may be involved in the migraine triggering process [7]. In any case, at brainstem level, as recently reviewed by Villalón and MaassenVanDenBrink [8], these mechanisms may involve: (i) low plasma concentrations of serotonin (5-hydroxytryptamine; 5-HT); (ii) high plasma concentrations of calcitonin gene-related peptide (CGRP); and (iii) vasodilatation of intracranial and extracranial arteries, which are a significant source of pain in migraine, seemingly associated with CGRP release from trigeminal sensory nerves. This CGRP-induced cranial vasodilatation, in turn, mechanically activates perivascular trigeminal sensory nerves resulting in release of neurotransmitters that promotes a peripheral inflammatory response within the dura and peripheral sensitization of trigeminal nerves.

CONTACT Carlos M. Villalón cvillalon@cinvestav.mx

Article highlights

- Migraine pathophysiology involves, amongst others, low plasma concentrations of serotonin (5-hydroxytryptamine; 5-HT) and high plasma concentrations of calcitonin gene-related peptide (CGRP).
- Consequently, the triptans ($5\text{-HT}_{1B/1D}$ receptor agonists), the ditans (5-HT_{1F} receptor agonists), the gepants (CGRP receptor antagonists) and human(ized) monoclonal antibodies against CGRP or its receptor have been developed as antimigraine drugs.
- However, the above drugs are not completely free of adverse side effects.
- For acute antimigraine therapy, the triptans, ergots and non-steroidal anti-inflammatory drugs (NSAIDs) are effective, but may produce medication overuse headache (MOH) and a wide variety of adverse (mainly cardiovascular) side effects.
- Lasmiditan (a 5-HT_{1F} receptor agonist) is devoid of vasoconstrictor properties (unlike the triptans), but its capability to cross the blood brain barrier (BBB) results is central side effects (*e.g.* nausea, somnolence, fatigue, vomiting, paresthesia, dizziness, etc.).
- Regarding the CGRP receptor antagonists and humanized monoclonal antibodies against CGRP or its receptor, the further development of some of the former was interrupted because they produced hepatotoxicity; whereas the latter are promising and apparently devoid of important adverse side effects, but long term (Phase III and IV) studies are required to definitely delineate their adverse side effect potential.

This box summarizes key points contained in the article.

1.2. A brief history of headache treatment

Some reviews [e.g. 8,9] have described a wide variety of methods used throughout the centuries (even millennia!) in an attempt to alleviate headache (including migraine). These have ranged from skull trepanation (8500 BC) to the use of a linen strip for collapsing distended temporal arteries which were causing the pain in an individual with clear migraine symptoms (as depicted in the Ebers Papyrus, 1500 BC). Subsequently: (i) Hippocrates was the first to describe the visual symptoms ('aura') of migraine-like headache (around 400 BC); (ii) Galen wrote of 'a painful disorder affecting approximately one-half of the head' and named it 'hemicrania' (2nd Century AD), which was gradually transmuted into 'migraine'; (iii) Thomas Willis published in 1664 that 'megrim' was due to dilatation of blood vessels within the head; (iv) William H. Thomson introduced in 1894 ergot extracts as an effective remedy for migraine, although physicians were aware of the intoxication risk (i.e. ergotism or St. Antony's Fire) when taken frequently; and (v) Arthur Stoll published in 1920 the isolation of ergotamine from the ergot fungus at Sandoz, a finding that represented the beginning of an effective and established antimigraine therapy. Significantly, in the late 1930s, Harold Wolff was the first to demonstrate pulsations in the temporal branches of the external carotid artery during migraine attacks and that pain could be relieved either by physical compression or by ergotamine. These (and other) findings reinforced the vascular theory of migraine and led to propose 5-HT as the vasoactive substance most likely involved in migraine because, amongst other reasons: (i) 5-HT concentrations in blood decrease during the headache phase and (ii) migraine attacks may be relieved by an i.v. infusion of 5-HT. These findings led to consider migraine headache as a 'low 5-HT syndrome'. Then, in the early 1990s, the neurogenic theory of migraine was strengthened when showing that trigeminovascular axons from blood vessels of the pia mater and dura mater release vasodilator neuropeptides, including CGRP, substance P, and neurokinin A, producing a sterile inflammatory reaction (neurogenic inflammation) with pain [8,9].

1.3. Current antimigraine drugs

The history of headache and migraine treatment (as described above) is undoubtedly fascinating and spans from the Neolithic period to the Space Age. Nonetheless, the number of effective antimigraine drugs had been, until recent times, limited. As reviewed by several authors [e.g. 4,9,10,11,12,13,14], the last decades have witnessed groundbreaking developments in antimigraine therapy. Currently, antimigraine drugs can be divided into: (i) acutely acting antimigraine drugs (i.e. agents that abolish an individual migraine attack; for example, ergots, triptans, and nonsteroidal anti-inflammatory drugs [NSAIDs]); and (ii) prophylactic antimigraine drugs (agents aimed at its prevention; e.g. β-adrenoceptor blockers, antiepileptics, Ca^{2+} channel blockers, 5-HT_2 receptor antagonists, monoclonal antibodies against CGRP or its receptor, etc.). Certainly, acute and prophylactic antimigraine drugs may also fall into two broad categories depending on their mechanisms of action, namely: (i) specific antimigraine drugs (i.e. ergots, triptans, ditans, gepants, and monoclonal antibodies against CGRP or its receptor) and (ii) nonspecific antimigraine drugs (i.e. NSAIDs, antiemetics, β-blockers, antiepileptics, etc.) [9]. Clearly, the prophylactic antimigraine armamentarium has a wider availability of antimigraine drugs than the acute armamentarium does [10]. Although most migraineurs need acute treatment, those who suffer from frequent attacks or have contraindications for acute treatment are also prescribed prophylactic drugs [15]. These antimigraine therapies have important differences in their selectivity, efficacy, adverse side effects (see Table 1), and cost [16,17]. For example, NSAIDs are very popular because they are cheap, effective, and easy to administer, but they also produce medication overuse headache (MOH) after long-term use, as also observed with the ergot alkaloids and the triptans (see Table 1).

Regarding specific antimigraine drugs, the triptans represent a top therapeutic breakthrough [11,26], but their vasoconstrictor side effects (see Table 1) warrant caution in patients with cardiovascular pathologies [9,27]. Other side effects such as nausea, dizziness, and chest symptoms preclude some patients from using triptans [27], while a number of patients does not respond well to the triptans. Certainly, compliance and tolerability of triptans is different for each drug (see Table 1 for references). This unmet need for better acute antimigraine treatments has led to the development of better acute antimigraine agents (see Figure 1), such as the ditans (5-HT_{1F} receptor agonists; i.e. lasmiditan), the gepants (CGRP receptor antagonists; i.e. telcagepant, MK-3207, etc.), and most recently, the human(ized) monoclonal antibodies targeting CGRP or its receptors (see below).

Table 1. Most common adverse side effects produced by the use of acute and prophylactic antimigraine drugs.

Acute pharmacotherapy				
Substance	Dosage (daily mg)	Level[a]	Side effects (>5%)	Comments
Ergot alkaloids				
Ergotamine tartrate Dihydroergotamine	2 (oral) 2 (suppository)	**A**	Nausea, vomiting, paresthesia, and ergotism	Given to patients with long-lasting migraine attacks. To prevent MOH, their use must be limited to 10 days/month. Contraindicated in patients with cardiovascular and cerebrovascular disease, Raynaud's disease, arterial hypertension, renal failure, pregnancy, and lactation.
Triptans				
Sumatriptan	25,50,100 (oral) 25 (suppository) 10, 20 (nasal spray) 6 (s.c.)	**A**	Chest discomfort, nausea, dizziness, distal paresthesia, fatigue	Although the triptans showed better efficacy than ergot alkaloids, their use should be restricted to 10 days/month to prevent MOH. Between 15% and 40% of patients with oral triptans experienced a worsening of headache after pain free. Contraindicated in patients with arterial hypertension, coronary heart disease, Raynaud's disease, history of ischemic stroke, pregnancy, and lactation.
Zolmitriptan	2.5, 5 (oral) 2.5, 5 (nasal spray)			
Naratriptan	2.5 (oral)			
Almotriptan	12.5 (oral)			
Rizatriptan	10 (oral)			
Eletriptan[b]	20, 40, 80 (oral)			
Frovatriptan	2.5 (oral)			
NSAIDs				
Acetylsalicylic acid	1000 (oral) 1000 (i.v.)	**A**	Gastrointestinal side effects. Risk of bleeding. Can trigger bronchospasm or anaphylaxis in asthmatic patients	To prevent MOH, their intake should be restricted to 15 days/month.
Ibuprofen	200–800			
Naproxen	500–1000			
Diclofenac	50–100			
Miscellaneous analgesics				
Paracetamol	1000 (oral)	**A**	Fewer adverse gastrointestinal effects (compared to NSAIDs). Caution in liver and kidney	To prevent MOH, their intake should be restricted to 15 days/month.
Metamizole	1000 (oral) 1000 (i.v.)	**B**	Agranulocytosis. Hypotension. Can trigger bronchospasm or anaphylaxis in asthmatic patients.	
Phenazone	1000 (oral)		Caution in liver and kidney	
Antiemetics				
Metoclopramide	10–20 (oral), 20 (suppository), 10 (i.m., i.v., s.c.)	**B**	Dyskinesia	Considering their extrapyramidal side effects, domperidone has less severe effects than metoclopramide and can be given to children (10 mg).
Domperidone	20–30 (oral),			

PROPHYLACTIC PHARMACOTHERAPY				
Substance	Dosage (daily mg)	Level[a]	Side Effects	Comments
β-blockers				
Metoprolol Propranolol	50–200 40–240	**A**	Fatigue, insomnia, dizziness, depression, postural hypotension, and bronchoconstriction	Although propranolol, valproate and topiramate reduce the frequency of migraine attacks (~50%) in 40–50% of patients, the treatment is not well tolerated (except for propranolol) due to their adverse side effects.
Ca^{2+} channel blockers				
Flunarizine	5–10	**A**	Weight gain, depression, drowsiness, and parkinsonism	
Antiepileptic drugs				
Valproic acid	500–1500	**A**	Sedation, nausea, weight gain, tremor, hair loss, liver toxicity, and teratogenicity	
Topiramate	25–100	**A**	Paresthesia, fatigue, dysgeusia, nausea, cognitive dysfunction, and depression	
Miscellaneous				
Amitryptiline	50–150	**B**	Weight gain, dry mouth, sedation, and drowsiness	
Venlafaxine	75–150		Dry mouth, sedation, insomnia, and fatigue	
Naproxen	2 x 250–500		Gastrointestinal side effects. Risk of bleeding.	
Bisoprolol	5–10		Fatigue, insomnia, dizziness, depression, postural hypotension, and bronchoconstriction	

It is noteworthy that the gepants, and antibodies against CGRP or its receptor represent new treatments under development and, currently, there are no conclusive randomized controlled clinical trials. The data presented here only show the most common adverse side effects of drugs with Level A or B of recommendation [18]. Acute antimigraine drugs are considered abortive and are related with: (i) agonism on serotonin $5\text{-}HT_{1B/1D/1F}$ receptors, (ii) prostaglandin synthesis inhibition, and (iii) dopamine D_2-like antagonism. Prophylactic drugs are preventive medications and are related with: (i) β-adrenoceptor antagonism, (ii) reuptake inhibition of noradrenaline and serotonin, (iii) inhibition of voltage-gated Na^+ and Ca^{2+} channels, (iii) $GABA_A$ receptor activation and GABA increases, (iv) $5\text{-}HT_2$ receptor antagonism, and (vi) prostaglandin synthesis inhibition.

MOH: medication overuse headache; NSAIDs: nonsteroidal anti-inflammatory drugs.

[a]Level A. Established as effective, requires at least two consistent Class I studies.

[b]Level B. Established as probably effective, requires at least one convincing Class I or two consistent Class II studies.

Local guidelines may differ between countries.

Data taken from [19–25].

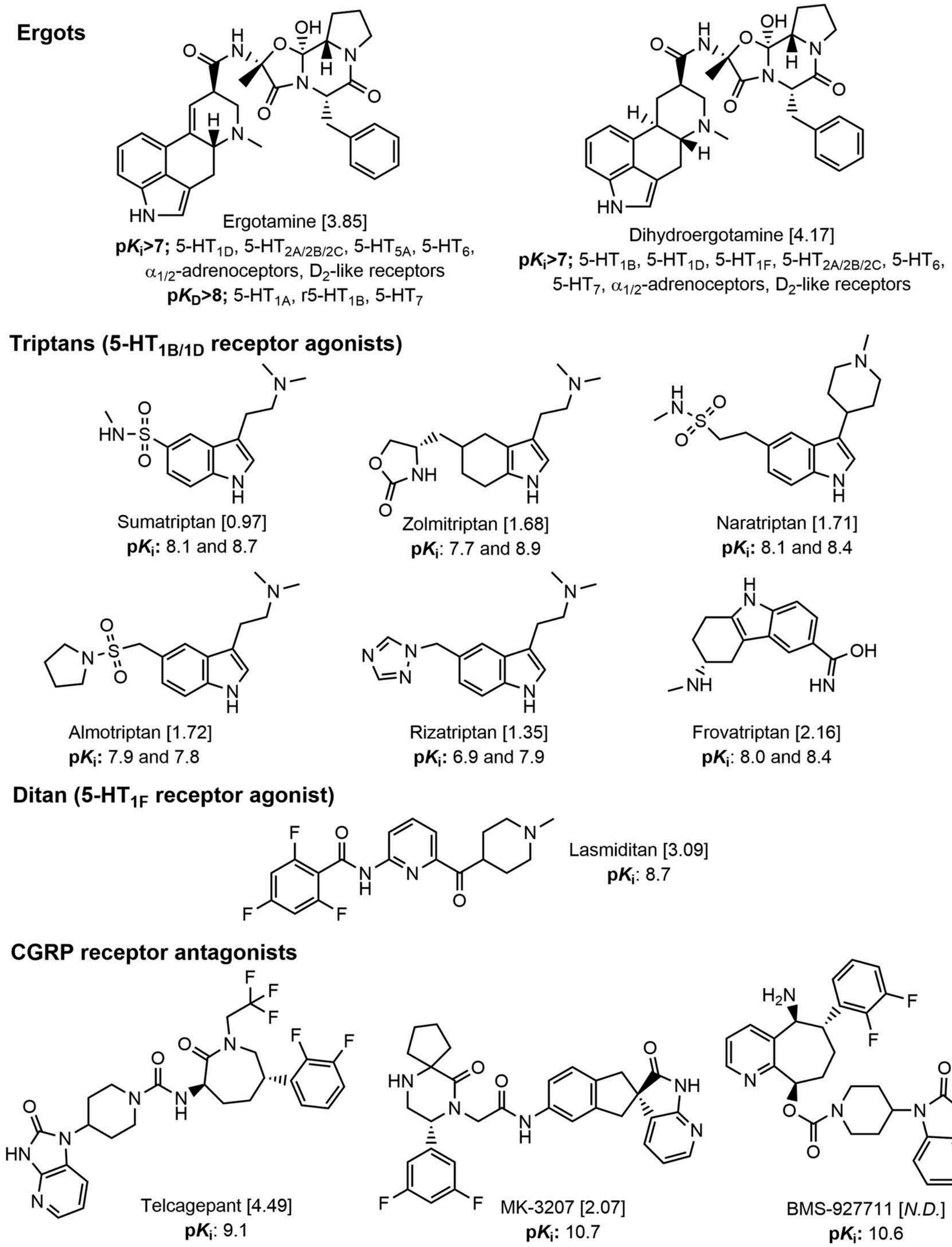

Figure 1. Chemical structures of some of the most relevant specific antimigraine drugs currently used or in development. Their corresponding affinity constants are presented as pK_i values (calculated in human tissues, except for the rat 5-HT$_{1B}$ receptor in the case of ergotamine). Moreover, their partition coefficients (indicators of liposolubility) are presented as Log p values [in brackets]. The ergots were the first specific antimigraine approach based primarily on their capability to produce cranial vasoconstriction; however, these compounds display affinity for several receptor types and also produce coronary vasoconstriction and other adverse side effects [7]. This fostered the search for more selective antimigraine drugs that led to the development of sumatriptan and second generation triptans, which were designed as selective vasoconstrictors acting on 5-HT$_{1B/1D}$ receptors [7]. Nevertheless, most triptans also display affinity for 5-HT$_{1F}$ receptors. Both, the ergots and the triptans also exert effects at trigeminal and supraspinal levels, inhibiting the CGRPergic neurotransmission. In an attempt to avoid triptans' vasoconstrictor properties, further research has culminated with the development of the ditans (i.e. the specific 5-HT$_{1F}$ receptor agonist lasmiditan) and the gepants (i.e. CGRP receptor antagonists such as telcagepant, MK-3207, etc.). To date, only the ergots and the triptans are currently available as specific antimigraine agents, whereas lasmiditan is under pharmaceutical development. The gepants telcagepant and MK-3207 were discontinued due to hepatotoxicity. Data taken from [28] and [29].

An important advantage of the gepants (Table 2) and the monoclonal antibodies against CGRP (i.e. the nezumabs such as galcanezumab, eptinezumab, and fremanezumab) or its receptor (i.e. the numab called erenumab) (Table 3) over the triptans is the absence of vasoconstrictor activity, possibly allowing the treatment of migraineurs with cardiovascular risk factors. However, the potential cardiovascular dangers resulting from abolishing the vasodilator actions of endogenous CGRP by these new antimigraine drugs must be adequately evaluated [13]. Meanwhile, the acute and

Table 2. CGRP receptor antagonists. All CGRP receptor antagonists have demonstrated efficacy in phase II and phase III studies in acute migraine attacks.

Gepant	% headache relief after 2 h	Adverse events (>5%)	Notes
Olcegepant or BIBN 4096BS (2.5 mg; i.v.)	44%; Phase II	Paresthesias	Although effective in alleviating migraine attacks up to 6 h, this molecule was discontinued due to poor oral bioavailability [124].
Telcagepant or MK-0974 (300 mg; oral)	49–55%; two Phase III trials	Fatigue, dizziness, dry mouth, and somnolence	Development interrupted due to hepatotoxicity concerns after repeated administration [126,127].
MK-3207 (200 mg; oral)	69%; Phase II	None	The overall incidence of adverse events was comparable to placebo. Suspended due to hepatotoxicity concerns [129].
BI 44370 TA (400 mg; oral)	56%; Phase II	None	No adverse events occurred with frequency of 5% or more and the reported adverse events were similar (in proportion) to placebo. Although in one subject an increased liver function was detected, some intrinsic characteristics of the patient made it difficult to draw a conclusion on this issue. Assessment of hepatic effects with repeated dosing are needed [130].
Rimegepant or BMS-927711 (300 mg; oral)	75%; Phase IIb	None	The overall incidence of adverse events was comparable to placebo. Pharmaceutical development was discontinued for unknown reasons [131].

Table 3. Humanized monoclonal antibodies against CGRP or the CGRP receptor in patients with episodic and/or chronic migraine.

Compound	Dose	Efficacy of treatment	Adverse events[a] (>5%)	Notes
Galcanezumab or LY2951742 (Phase II)	150 mg (s.c.) given once every 2 weeks for 12 weeks	Better than placebo	Injection site pain and/or erythema, upper respiratory tract infections, and abdominal pain	This study was performed in subjects with 4–14 migraine headache days/month according to ICHD-2. Preliminary Phase III trials showed similar results to the Phase II trial [30].
Eptinezumab or ALD403 (Phase II)	Single dose, 1000 mg (i.v.)	Better than placebo	Upper respiratory tract infection, urinary tract infection, fatigue, back pain, and arthralgia	This study was performed in subjects with 5–14 migraine headache days/month according to ICHD-2. Under Phase III trial [31].
Fremanezumab or TEV-48125 (Phase IIb)	225 mg (s.c.) given monthly for 3 months	Better than placebo	Injection site pain and sinusitis	This study was performed in subjects with 8 migraine headache days/month, but not more than 14 headache-days of any phenotype (migraine or not) according to ICHD-3 beta version. Under Phase III trial [32].
Fremanezumab or TEV-48125 (Phase IIb)	675/225 or 900 mg (s.c.) given monthly for 3 months	Better than placebo	Injection site pain and/or pruritus, urinary tract infection	This study was performed in subjects diagnosed with chronic migraine defined as headaches for at least 15 days/month with at least 8 days of migraine/month according to ICHD-3 beta version. Under Phase III trial [33,34].
Erenumab or AMG 334 (Phase IIb)	70 mg (s.c.) monthly for 3 months	Better than placebo	Nasopharyngitis, fatigue and headache. 3% of patients receiving AMG 332 developed neutralizing antibodies	This study was performed in subjects with a history of migraine (with or without aura) for 12 months or longer according to ICHD-II and with 4–14 migraine days/month. Under Phase III trial [136].

[a]In all cases, the adverse events reported in the treatment group were similar to those observed in the placebo group. Although serious adverse events were also reported in all RCT, these seem to be unrelated to the drug under study. Furthermore, it has to be kept in mind that the number of patients could be insufficient to detect rare or uncommon serious adverse events.

prophylactic antimigraine drugs commercially available thus far (see Table 1), particularly those interacting with 5-HT receptors, remain up to date the most reliable election for the treatment of migraine without serious side effects.

2. An overview on the mechanisms of current and prospective antimigraine pharmacotherapies (acute and prophylactic)

2.1. General considerations

The currently available antimigraine drugs (both acute and prophylactic) have different mechanisms of action (see below). Nevertheless, they also have shortcomings and may cause adverse (including cardiovascular) side effects (see Table 1). These – and other – reasons (see below) have led to the development of antimigraine drugs devoid of vasoconstrictor effects. In this respect, several compounds with actions in the central nervous system (CNS) have been shown to inhibit neuronal transmission by interacting with CGRP receptors, glutamate receptors, VPAC/PAC receptors, NO synthase, 5-HT_{1F} receptors, and gap junctions [12]. These interactions producing inhibition of (or interference with) neuronal transmission may translate into acute or prophylactic antimigraine action, but also (directly or indirectly) into vascular (side) effects.

As pointed out in Section 1, augmented plasma concentrations of CGRP are a key component of the pathophysiology of migraine so that inhibition of trigeminal release of CGRP or interference with CGRP receptor activation may result in antimigraine action [4,13]. In contrast with the classical antimigraine drugs that were directed at inducing cranial vasoconstriction (ergots, triptans) or analgesia (NSAIDs, including paracetamol), the newest antimigraine drugs are directed at inhibiting CGRP release (lasmiditan, in addition to its direct central action) or interfering with CGRP receptor activation (gepants, monoclonal antibodies) in the trigeminovascular system; this, in turn, would result in prevention of both trigeminal vasodilatation and pain integration [35]. Preliminary studies with these new drugs have shown promising results in aborting and preventing migraine attacks [36].

2.1.1. Current acutely acting antimigraine drugs

These include:

(a) Ergot alkaloids (see Figure 1), which produce cranial and systemic vasoconstriction and can interact with serotonin 5-$HT_{1/2}$, dopamine D_2-like, and α_1/α_2-adrenoceptors [9].
(b) Triptans (see Figure 1), which produce a more selective cranial vasoconstriction than the ergots and are selective 5-$HT_{1B/1D}$ receptor agonists (and some can also interact with 5-HT_{1F} receptors) [16].
(c) Over-the-counter medicines (mainly NSAIDs), whose peripheral mechanisms of action, as previously described [37], are related to: (i) inhibition of cyclooxygenase (COX-1 and COX-2) and lipoxygenase enzymes, and (ii) disruption of signal transduction by uncoupling the G-protein-regulated processes. Furthermore, central effects of NSAIDs (particularly for indomethacin, ibuprofen, and diclofenac) have also been observed. In addition, miscellaneous analgesics (e.g. paracetamol, metamizole, and phenazone) and antiemetics (e.g. metoclopramide and domperidone) seem to be useful to treat acute migraine attacks (see Table 1 for details). Indeed, antiemetics can inhibit the trigeminovascular transmission [38].

2.1.2. Current prophylactic antimigraine drugs

A wide variety of medications not originally developed for antimigraine action have shown prophylactic antimigraine utility; these include β-blockers, antidepressants, antiepileptics, Ca^{2+} channel blockers, reuptake inhibitors of noradrenaline and 5-HT, etc. (see Table 1 for references). Nevertheless, the specific mechanisms of therapeutic action (regarding cephalalgias) remain speculative in most cases (see Section 2.3 below). A concern on prophylactic antimigraine drugs is the possibility to induce pivotal effects when withdrawn, and hence they might boost the occurrence or intensity of migraine attacks. Although prophylactic antimigraine drugs are not devoid of side effects (see Table 1), many studies suggest that an appropriate prophylactic treatment may represent an important improvement in migraineurs' quality of life. In addition, it is worthy of note that prospective prophylactic specific antimigraine drugs involve monoclonal antibodies targeting CGRP (e.g. galcanezumab, eptinezumab, fremanezumab) or its receptor (erenumab) directed to preventing migraine attacks with an effectiveness similar to that of other prophylactic antimigraine drugs, but with a hypothetical decrease in the induction of adverse side effects [39] (see Section 3).

2.2. Specific antimigraine drugs

2.2.1. Ergot alkaloids

Both ergotamine and dihydroergotamine (DHE) share structural similarities with serotonin, dopamine, and (nor)adrenaline, and have been shown to display affinity for several receptors including serotonin 5-$HT_{1/2}$, dopamine D_2-like, and α_1/α_2-adrenoceptors [9,40]. Ergotamine has a complex pharmacological profile [9,40] and its therapeutic antimigraine action involves mainly activation of 5-HT_1 receptors, specifically the 5-HT_{1B}, 5-HT_{1D}, and 5-HT_{1F} subtypes [16] (see Figure 2). Since serotonergic mechanisms play a key role in the modulation of CGRP release in the trigeminal system, an antinociceptive action of these ergots has been implied [41]. In addition, Valdivia et al. [42] showed that 5-HT_{1B}, $\alpha_{2A/2C}$-adrenoceptors and α_1-adrenoceptors mediate the canine external carotid vasoconstriction to ergotamine. In an attempt to reduce adverse (including cardiovascular) side effects, the molecule of ergotamine was modified to DHE by reducing an unsaturated bond in ergotamine (see Figure 1).

Indeed, DHE showed efficacy to abort migraine attacks with a much lower vasoconstrictor and emetic potential than ergotamine [40]. DHE has at least three main therapeutic actions to abort migraine attacks: (i) a vasoconstrictor action mainly on cranial blood vessels (via 5-HT_1 receptors); (ii) a decrease in the release of CGRP and other vasoactive neuropeptides (e.g. substance P and neurokinin A) via α_2-adrenoceptors, 5-HT_{1B}, and D_2-like receptors; and (iii) inhibition of the trigeminal pain integration (via α_2-adrenoceptors, 5-$HT_{1B/1D}$, and D_2-like receptors). In addition, intrathecal injections of DHE on the canine trigeminal nucleus inhibited the carotid vasodilatation to capsaicin via 5-$HT_{1B/1D}$ receptors and α_{2C}-adrenoceptors [43,44]; these findings suggest also a potential spinal mechanism for DHE (i.e. inhibition of spinal reflexes). It is noteworthy that ergotamine and DHE are still in use as antimigraine drugs when migraineurs suffer from low frequency attacks. However, even when both drugs are useful, they have important cardiovascular side effects that limit their use and, consequently, would be contraindicated in patients with cardiovascular and cerebrovascular disease, arterial hypertension, and renal failure (see Table 1). In addition to their acute antimigraine actions, some clinical assays have shown certain prophylactic effect using a time-release preparation of DHE [45].

2.2.2. Triptans

After ergots' age, still under the vascular theory of migraine, the triptans (5-$HT_{1B/1D}$ receptor agonists) were developed as selective vasoconstrictors of cranial blood vessels in order to diminish cardiovascular side effects [11]. The first triptan developed by Glaxo and launched to the market in 1991 was sumatriptan (GR43175). Although certainly effective in the acute treatment of migraine, particularly when given parenterally, sumatriptan's low oral bioavailability led to the development of the second-generation triptans [see Figure 1; for references see 9]. In general, triptans' mechanisms of action in the trigeminovascular system may include: (i) cranial vasoconstriction via 5-HT_{1B} receptors [9,46], (ii) inhibition of trigeminal CGRP release [16], and (iii) inhibition of trigeminal pain transmission [47].

Currently, the triptans are still considered as first choice drugs to specifically abort migraine attacks. These compounds are certainly more selective than the ergots to induce cranial vasoconstriction, but they are not devoid of cardiovascular side effects [48]; consequently, they are contraindicated in patients with cardiovascular pathologies (see Table 1), as they have the propensity to produce coronary vasoconstriction [49].

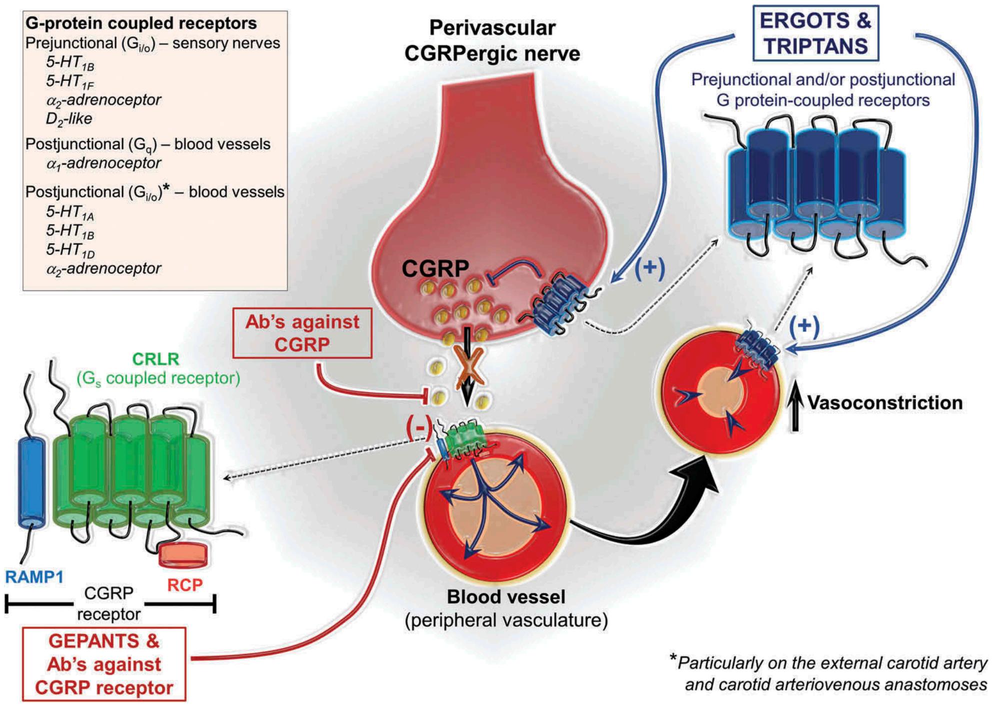

Figure 2. An overview of mechanisms/actions of antimigraine drugs on perivascular CGRPergic transmission. CGRP released from sensory nerves exerts predominantly a vasodilator action by activation of vascular CGRP receptors. The CGRP receptor is an atypical G-protein coupled receptor (GPCR) that requires additional complementary proteins to be functional, namely RAMP1 and RCP. Interestingly, the ergots and the triptans -apart from their established vasoconstrictor (postjunctional) effects [6,7]-, can modulate perivascular CGRPergic transmission; in brief, these compounds may inhibit CGRP release by prejunctional activation of GPCR receptors coupled to $G_{i/o}$ proteins (i.e. $5\text{-}HT_{1B/1F}$ for the triptans as well as α_2-adrenoceptors and D_2-like receptors for the ergots) (for refs. see [138]). Although the development of most gepants (CGRP receptor antagonists) was interrupted due to hepatotoxicity, the development of humanized monoclonal antibodies (*Ab*'s) targeting CGRP or the CGRP receptor is promising. The mechanisms depicted here are present at spinal (on the trigeminocervical complex) and supraspinal levels where the CGRPergic transmission is also present between a presynaptic and postsynaptic neuron.

Like the prophylactic antimigraine drugs, the triptans seem to be useful in certain types of migraine. For example, Moschiano et al. [50] reported that naratriptan decreased the number of pure menstrual migraine attacks when given 2 days before expected menstrual onset and continuing for 6 days; however, overuse may produce MOH. Furthermore, a meta-analysis suggests that the acute short-term effective prophylactic treatment using frovatriptan is better than that observed with the other triptans [51], an effect probably related with the very long plasma elimination half-life ($t_{1/2}$ ~ 26 h).

Finally, it is worthy of note that even when ergots and triptans are very useful therapeutic tools to treat migraine, only around 30% of migraineurs are relieved without important side effects. One explanation for this was suggested by Christensen et al. [52], who found a correlation between genetic constitution (single nucleotide polymorphisms) and responses to classic antimigraine pharmacotherapies. The above urges: (i) the obligation of personalizing the clinical practice according to real necessities of each patient and/or (ii) to develop other alternatives for acute antimigraine pharmacotherapy.

2.2.3. Ditans

The cardiovascular side effect potential of the triptans (of which some can also activate $5\text{-}HT_{1F}$ receptors without producing vasoconstriction) led to the development of selective $5\text{-}HT_{1F}$ receptor agonists (also called ditans) [9,11]. This culminated with clinical studies showing that the lipophilic $5\text{-}HT_{1F}$ receptor agonist, lasmiditan (see Figure 1), was effective in the acute treatment of migraine without producing cardiovascular side effects, as compared to placebo [53,54]. However, some CNS adverse side effects were also observed (see Section 3 below).

2.2.4. Gepants and antibodies against CGRP or its receptor

Among newest and highly selective antimigraine pharmacotherapies, one gepant, nezumabs and numabs are under clinical development with promising results. Gepants are CGRP receptor antagonists that were developed to prevent the interaction of CGRP with its receptor into the trigeminovascular system (see Table 2). It is not yet completely clear whether gepants' site of action is located in the periphery, in the CNS or in both [55]. Nevertheless, gepants' mechanisms of action to abort migraine attacks may involve: (i) prevention of cranial vasodilatation induced by CGRP [56] and (ii) blockade of trigeminal pain transmission, at central [26] or peripheral (at trigeminal ganglia) [57] level. On the other hand, recent results reported by Walker et al. [58] have suggested the existence of a second receptor responsive to CGRP. Hence, the possibility of a novel site of action for the gepants remains plausible and definitely requires further research.

Importantly, the gepants have an efficacy to abort migraine attacks comparable to that of the triptans (without producing vasoconstriction; for references see [4]). However, these compounds may exert some important liver toxicity effects when given (sub)chronically, although this does not seem to be a class effect (see Section 3). This has led to a novel strategy that has culminated in the development of human(ized) monoclonal antibodies, which target CGRP or its receptor (see Table 3).

These human(ized) monoclonal antibodies are protein-based macromolecules designed to recognize, bind, and disable (i.e. scavengers) the CGRP molecule or its receptor. One advantage of these antibodies over the gepants would be their longer half-life [13] and a lower hepatic impact as compared with classic drugs [59]. Furthermore, since their large molecular size should be an impediment to cross the blood–brain barrier (BBB), minimal central side effects could be induced. Indeed, these antibodies have been analyzed in randomized clinical trials (RCT) with the promise of better tolerance [14]. The main mechanisms of action of the antibodies against CGRP or its receptor remain unknown, but inhibition of CGRPergic transmission at trigeminovascular level seems to be involved, resulting in an inhibition of both nociceptive transmission and cranial vasodilation.

2.3. Nonspecific antimigraine drugs

NSAIDs are highly prescribed by general practitioners to treat migraine and have shown certain usefulness to reduce the intensity of headache during acute migraine attacks [60]. Lamentably, the efficacy of NSAIDs seems to be limited to low or moderate frequency of migraine attacks since, as observed with the triptans, MOH after overuse could be induced [61].

The β-adrenoceptor blocker propranolol, which is frequently used as a prophylactic antimigraine drug [62], has been suggested to exert its therapeutic action by interrupting nociceptive transmission at spinal and supraspinal levels [63]. Certainly, propranolol reduces trigeminovascular responses in the thalamic ventroposteromedial nucleus [64], may block serotonin receptors [65], inhibits CSD by blocking glutamate release [66], and stabilizes the inactivated state of Na^+ channels [67].

Topiramate is a classic anticonvulsant that is clinically prescribed to prevent chronic migraine. Some evidence showed beneficial effects in adults [68], whereas in pediatric patients opposite results have been reported [69]. About its mechanisms of action, this compound enhances GABAergic transmission and block glutamate receptors and/or voltage activated Na^+ and Ca^{2+} channels [70].

The antiepileptic sodium valproate seems to be effective in aborting [71] and preventing [72] migraine attacks. Its mechanisms of action seem to include: (i) an increase in GABA levels because of multiple interactions with GABA transporters and enzymes involved in GABA metabolism, (ii) a decrease in excitatory components of the CNS (e.g. aspartate levels), and (iii) neuroprotection against oxidative stress induced by NMDA via tropomyosin receptor kinase B (TrkB).

3. Side effects associated with antimigraine pharmacotherapies

3.1. Overview of common and specific adverse side effects

As previously pointed out, the armamentarium of antimigraine drugs has been enriched with the incorporation of relatively more selective agents during the last decades [4,10,13]. Nevertheless, the search for new antimigraine drugs remains a challenge, in part because not all patients respond positively to treatment, and a wide variety of adverse side effects (see Table 1 and Figure 3). The side effects are miscellaneous (Figure 4) and associated not only with the intrinsic mechanisms of action, but also with liver and kidney toxicity, particularly relevant for metamizole and paracetamol [73], which are nonspecific antimigraine drugs. More recently, besides their gastrointestinal adverse side effects, NSAIDs (nonselective cyclooxygenase [COX] inhibitors) have been put on scrutiny since they may increase the chance of heart attacks or stroke [74]; this serious side effects are attributed to COX-2 inhibition in the cardiovascular system [75]. In addition, frequent intake of acute headache medications may induce MOH. Certainly, safety and tolerability of NSAIDs is quite different in different groups.

In the case of ergots, apart from vasoconstriction of intracranial and extracranial blood vessels at therapeutic doses, these compounds (mainly ergotamine) may induce nausea, vomiting, vertigo, restlessness (CNS effects), dry mouth, gastric, and chest symptoms [79], whereas an overdose or chronic overuse may precipitate MOH, ergotism as well as heart and brain infarction [80]. In humans, blood pressure is increased for about 3 h after a parenteral therapeutic dose of ergotamine or DHE [81]. This array of side effects is due not only to the capability of ergots to interact with $5\text{-HT}_{1/2}$ receptors, $\alpha_{1/2}$-adrenoceptors, and D_2-like receptors [9,40], which are ubiquitously expressed, but also because they: (i) are lipophilic molecules (see their partition coefficients in Figure 1) and (ii) possess a slow dissociation from the receptor [49].

The advent of triptans ($5\text{-HT}_{1B/1D(/1F)}$ receptor agonists) was a major improvement in the acute migraine treatment since fewer side effects were observed (as compared to ergots) [17], although some isolated cases of probable toxicity have been reported, namely: (i) a case of sulfhemoglobinemia induced by sumatriptan [82] and (ii) a case of hepatotoxicity by rizatriptan, mediated by increasing oxidative stress through lysosomal and mitochondrial dysfunction in hepatocytes [83].

Furthermore, although sumatriptan (1st generation) is less lipophilic (Figure 1), the headache relief obtained is similar to that produced by the more liposoluble triptans (2nd generation) (Figure 3). In any case, considering that 5-HT_{1B}, 5-HT_{1D}, and 5-HT_{1F} receptors are expressed in several areas in the CNS [84,85], it is reasonable to suppose that the triptans could induce undesirable central effects depending on their liposolubility. Certainly, sumatriptan may induce central effects, but the occurrence in the population is low [86,87].

The correlation between liposolubility and CNS effects is clearly exemplified in a Phase II RCT with the use of the lipophilic 5-HT_{1F} receptor agonist, lasmiditan [53], where an increase of CNS adverse side effects was observed

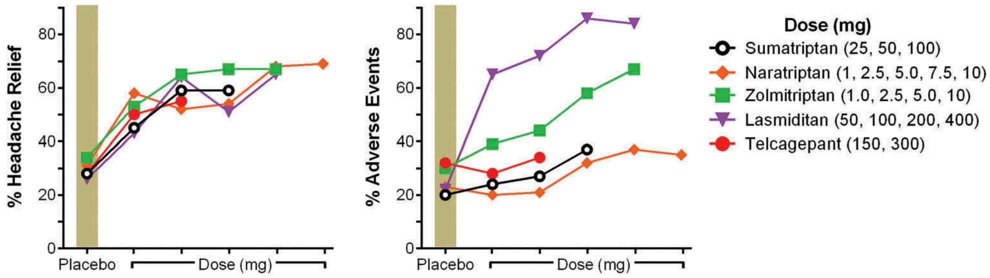

Figure 3. An overview of headache relief (at 2 h; acute treatment) and some adverse effects reported in randomized clinical trials (RCT) displayed for sumatriptan (1003 patients, Phase III trial), naratriptan (643 patients, Phase II trial), zolmitriptan (1144 patients, Phase III trial), lasmiditan (391 patients, Phase II trial) and telcagepant (1901 patients, Phase III trial). These RCT were realized with compounds taken orally and compared against its placebo effect. Interestingly, when extracting the placebo effect only between ~25–40% population had headache relief! Furthermore, though lasmiditan was devoid of vasoconstrictor effects, the proportion to alleviate migraine headache is similar to sumatriptan; unfortunately, the number of adverse side effects (mainly in the CNS) is high. Regarding telcagepant, although its pharmaceutical development was interrupted, its adverse side effects were similar to placebo. Data taken from [53,76–78,126].

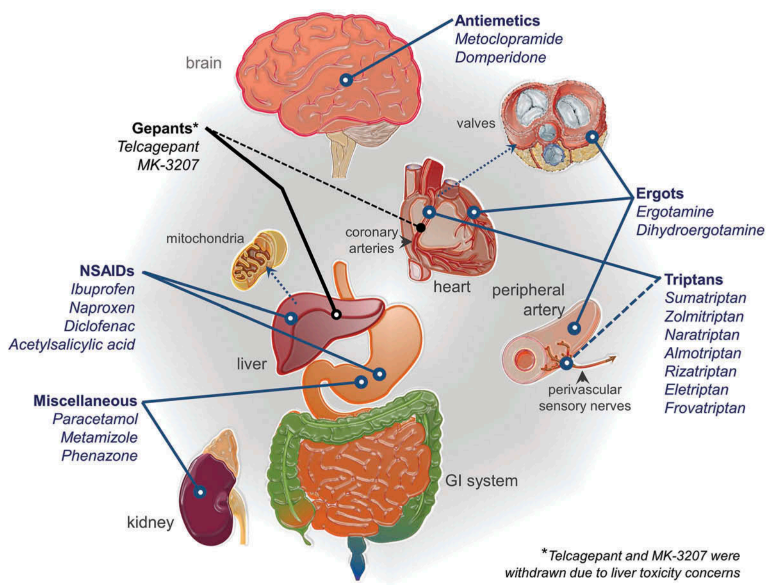

Figure 4. An overview of the adverse side effects observed with current migraine pharmacotherapy. In the case of the ergots and the triptans (specific antimigraine drugs), cardiovascular side effects have been the 'Achilles' heel'. Certainly, the ergots produce more side effects than the triptans, in part because the ergots – being less selective- display affinity for more receptors. The ergots and the triptans can induce direct vasoconstriction and inhibition of CGRPergic neurotransmission at peripheral blood vessels. Furthermore, a probable inhibition of CGRPergic systemic transmission has been observed for sumatriptan. Similarly, current data of gepants show that their acute administration has no direct vasoconstrictor effects. In the case of non-steroidal anti-inflammatory analgesics (NSAIDs), the most common adverse effects have been related with hepatotoxicity and gastrointestinal (GI) side effects, whereas paracetamol, metamizole and phenazone may produce GI and kidney side effects.

(dizziness > 23%, nausea > 5%, somnolence > 8%, fatigue > 10%, and sensation of heaviness > 5%). At this point, we need to stress that 5-HT_{1F} receptors seem to be present in cortical areas, hippocampal formation, and claustrum, regions that are meager of 5-$HT_{1B/1D}$ receptors [84,85]; this discrete localization, which may be influencing the increase of adverse side effects, should be taken into account when used in the clinical practice. Nevertheless, lasmiditan is devoid of vasoconstrictor effects in human blood vessels [88], a property that confers a cardiovascular safety advantage which is a main issue on triptans' therapy (see below).

Nevertheless, we must keep in mind that, apart from drugs' liposolubility, the integrity of BBB during migraine attacks may play a role [89,90]. Indeed, some authors suggest that during migraine attacks the BBB is disrupted; consequently, the antimigraine drugs easily penetrate the CNS (independently of their liposolubility) exerting their therapeutic action mainly by central mechanisms. In this respect, two studies using triptans strongly support this notion [91,92]. Briefly, comparing the effect of lipophilic (donitriptan or zolmitriptan) and nonlipophilic (sumatriptan) molecules (given parenterally) on well-validated acute migraine animal models where the BBB's are intact, only donitriptan and/or zolmitriptan inhibited: (i) c-fos expression at Sp5c/$C_{1\text{-}2}$ level induced by electrical stimulation of the superior sagittal sinus (SSS) [91] and (ii) capsaicin-induced vasodilatation of the canine external carotid circulation [92]. Accordingly, inhibition of Sp5c/$C_{1\text{-}2}$ neurotransmission by sumatriptan was only observed when the BBB was artificially disrupted using mannitol [93] or when given spinally [94].

In contrast, studies in humans using positron emission tomography (PET) during migraine attacks failed to show a decrease in brainstem neuronal activity after a therapeutic dose of subcutaneous sumatriptan [95]. Similarly, when migraine attacks were induced by glyceryl trinitrate, ^{11}C-dihydroergotamine (given intravenously) failed to cross the BBB [96]; interestingly, dihydroergotamine has a high partition coefficient (see Figure 1), but its capability to penetrate the brainstem is low due to its large molecular size [97]. Most recently, using resonance imaging in humans, Hougaard et al. [98] showed that the BBB remains intact during migraine with aura. Together, these studies in humans argue against the hypothesis supporting a disruption of the BBB during migraine attacks. Nevertheless, considering the limited spatial resolution of PET, we cannot exclude a possible effect of sumatriptan and dihydroergotamine in areas close-related to the cerebrospinal fluid (i.e. periaqueductal or the area postrema). Certainly, as noted by Tfelt-Hansen [90], CNS adverse events of sumatriptan suggest that this triptan could cross the BBB and probably a supraspinal therapeutic effect cannot be categorically excluded, as suggested by Weiller et al. [95].

3.2. Vascular and coronary side effects of ergots and triptans

Although ergot alkaloids are cranial vasoconstrictors (an effect probably partly responsible for their therapeutic effect [26]), they are also capable of exerting some unspecific peripheral vasoconstriction. These actions on peripheral blood vessels limit their use and are contraindicated in patients at risk for cardiovascular complications or abnormal blood pressure [27]. The main effect of ergotamine and DHE in the periphery is arterial vasoconstriction, but DHE has fewer side effects than ergotamine [40]. At therapeutic concentrations, these ergots induce a potent vasoconstriction in the canine external carotid vascular bed by activation of α_2-adrenoceptors and 5-HT$_{1B}$ receptors [42,99]. Ergotamine induces a more potent vasoconstriction on peripheral arteries (e.g. pulmonary and coronary arteries) [49,100], whereas DHE predominantly affects the capacitance blood vessels [40].

In the case of triptans, *in vitro* experiments showed that sumatriptan is predominantly cranioselective [101], but it also contracted the human proximal [49,102] and smaller distal [101] coronary arteries. These proximal and distal coronary arteries are involved in angina-like symptoms and myocardial ischemia. Nevertheless, as stated by MaassenVanDenBrink et al. [49], 'in view of low efficacy (in human coronary arteries) these drugs are unlikely to cause myocardial ischemia at therapeutic plasma concentrations in healthy humans'. However, in humans, after oral intake of sumatriptan (50 mg), rizatriptan (10 mg), or zolmitriptan (2.5 mg) at therapeutic dosages, the mean arterial pressure was increased by 4–6% [103] and the buffering capacity of conduit arteries was decreased [104]. In any case, in an open-label prospective study of 12,339 patients (excluding any subject with cardiac disease) O'Quinn et al. [86] suggested that sumatriptan seems not to increase per se the incidence of cardiovascular events. Nonetheless, cardiac death events have been reported after sumatriptan intake with a prevalence of 1 per 2,500,000 migraine attacks (in 9 million patients) [87]. Furthermore, in these 9 million patients, 10 cerebrovascular fatalities have been reported. These results contrast with a prospective study analyzing the incidence of sudden cardiac death in the general population, which ranges from 50 to 100 per 100,000 subjects [105].

3.3. Medication overuse headache

Frequent use of abortive antimigraine drugs may induce MOH (see Table 1), which represents a challenge to headache management. This disease is considered the most common cause of chronic daily headache from an episodic primary headache [106,107]. In general, the prevalence of MOH ranges from 0.5% to 2.6 % in the population [108] and 11–70% among patients with chronic daily headache [109]. This phenomenon is characterized by worsened or more frequent headache caused by regular medication intake [110]. Moreover, it is interesting to note that: (i) triptans' overuse seems to cause MOH more frequently than ergots do, and they seem to induce migraine-like headache rather than tension-type headache as observed with other medications and (ii) MOH induced by triptans is faster and at lower doses than that induced by other groups of drugs [111].

Although the mechanisms that lead to MOH are unknown, the increase of headache frequency, expansion of headache area, and cutaneous allodynia implies a central sensitization of the trigeminal pathways [112]. Basic research in rodents showed that chronic medication with triptans not only increases CGRP expression and neuronal nitric oxide synthase (nNOS) at trigeminal ganglia [113,114], but also increases the excitability of cortical neurons leading to cortical spreading depression (CSD) [115]. Indeed, MRI in humans with MOH showed a dysfunction of cortical and subcortical regions involved in pain processing [116]. It is noteworthy that although several classes of drugs could induce MOH, the clinical features are similar, suggesting some common shared mechanisms unrelated to their direct pharmacological effects [117]. MOH also shares some pathogenic mechanisms and structures with drug addiction (e.g. physical dependency to ergots) [116–118].

All classes of symptomatic medications (ergots, triptans, NSAID's) can cause MOH (see Table 1), and the relevance of an effective antimigraine drug regimen cannot be overstated considering that repeated attempts to abort a migraine attack may result in overmedication and, consequently, in MOH. Currently, the discontinuation or withdrawal therapy remains as the only treatment for this disorder [107].

3.4. Potential adverse side effects produced by gepants and antibodies against CGRP or its receptor

3.4.1. General

As previously described, a new generation of antimigraine drugs has emerged based on the premise that the CGRPergic system at trigeminovascular level plays a key role in the pathophysiology of migraine [4,13]. Furthermore, the development of these molecules has been fostered in part due to the cardiovascular issues observed with ergots and triptans. Indeed, a probable advantage of using CGRP receptor antagonists is the fact that these compounds (i.e. olcegepant) do not produce vasoconstrictor effects in human cranial and coronary arteries [119,120]. Most importantly, in patients with coronary artery disease, oral telcagepant (300 mg) seems to be secure since no episodes of chest pain were reported during treatment [121], but the effects of chronic CGRP blockade remain unknown. Nevertheless, as recently considered [13], studying patients with stable coronary artery disease due to atherosclerosis is not the most appropriate model to assess the safety profile of this new rationale (i.e. blockade of CGRP transmission) against migraine. In any case, this led to the development of CGRP receptor antagonists (gepants) and human(ized) monoclonal antibodies against CGRP or its receptor (see Tables 2 and 3). Targeting the CGRP system seems to be a good idea, but CGRP is involved in several physiological functions and not only in cardiovascular function; accordingly, this represents the potential for adverse side effects. Indeed, CGRP receptors are found in the CNS as well as in the peripheral nervous system (i.e. blood vessels, adrenals, bone, kidney, pancreas, etc.) [122].

In a first attempt (phase I and II trials) to prove the clinical relevance of blocking CGRPergic transmission to abort migraine attacks, i.v. olcegepant (BIBN4096BS) showed good safety and tolerability with few adverse side effects (see Figure 3) [123,124]. These data were promising and orally bioavailable gepants were designed (i.e. telcagepant [MK-0974] and MK-3207). Moreover, in a Phase IIb adaptive design trial [125] and in two Phase III trials [126,127], telcagepant seemed to be effective in the acute treatment of moderate or severe migraine attacks. However, the recurrent use of this compound, twice daily for 3 months (intended to be a prophylactic drug) in a Phase IIa exploratory study trial unmasked a potential hepatotoxicity effect since an increase of liver transaminases was detected in some participants [128]. Similar effects in the liver were observed during phase I and II trials with MK-3207 [128,129]; consequently, the pharmaceutical development of these compounds was discontinued.

Furthermore, two additional oral CGRP receptor antagonists (BI 44370 TA and BMS-927711) have been developed and tested on RCT [130,131]; in both cases, general adverse effects were lower than those previously observed with gepants. Interestingly, BMS-927711 is devoid of effects on liver transaminases, whereas an increase of alanine transaminase, aspartate transaminase, and gamma glutamyl transpeptidase was observed in one subject 5 days after a single oral administration of BI 443370 TA (400 mg). In any case, it is noteworthy that, apart from hepatotoxicity concerns, a lower incidence of adverse effects was observed for all gepants (see Table 2).

Most recently, the potential use of antibodies targeting directly the CGRPergic transmission has been developed. These CGRP 'scavengers' are human(ized) monoclonal antibodies against the CGRP molecule or its receptor and possess a long half-life. All Phase II RCT (see Table 3 for details) showed that these molecules are superior to placebo in treating episodic or chronic migraine headache, and the adverse side effects reported were similar to their respective placebo groups. Although their site of action remains unknown (peripheral and/or central?), these molecules (which are not liposoluble) may have a predominant peripheral action considering that antibodies could not cross the BBB and, therefore, would not affect central CGRP actions. This idea is valid, although: (i) large protein molecules, such as insulin, can cross the BBB [132]; and (ii) classical non-lipophilic drugs (e.g. gabapentin [133]) can cross the BBB mimicking an endogenous substrate. Nevertheless, given the much greater molecular weight of antibodies (around 150,000 Da), they are unlikely to cross the BBB to a major extent [11]. Admittedly, further studies are required to specifically investigate whether these antibodies can cross to the BBB, as CGRP receptors are highly expressed in some central regions (e.g. cerebellum). Certainly, this new type of drugs with a long half-life (~45 days for 300 mg fremanezumab given i.v. [134].) could be potentially dangerous.

Currently, four monoclonal antibodies against CGRP are under Phase III RCT (NCT02797951 for galcanezumab; NCT02974153 for eptinezumab; NCT02945046 for fremanezumab; and NCT02483585 for erenumab). Moreover, one monoclonal antibody against the CGRP receptor, namely, erenumab (AMG 334), has been suggested to display a safety profile similar to that of its placebo [135], as previously reported in a Phase II trial [136].

3.4.2. *Disruption of the systemic cardiovascular CGRPergic neurotransmission*

It has to be kept in mind that CGRP contained in perivascular sensory afferent fibers is relevant to maintain cardiovascular homeostasis. Indeed, the systemic vascular tone is modulated by sympathetic and primary sensory [137] perivascular nerves via the release of, respectively, the vasoconstrictor noradrenaline and the vasodilator CGRP. Accordingly, CGRP seems to play a key role in some vascular diseases including hypertension, preeclampsia, and cerebral ischemia; in all cases, CGRP is functioning as a physiological protector inducing vasodilatation [see 13, 138]. In addition, at cardiac level, CGRP acts against ischemia by inducing vasodilatation and improving myocardial contractility (for details see [13]).

A study in anesthetized rats [139] showed that i.v. olcegepant (3 mg/kg), which blocked the vasodepressor responses to CGRP, had no effect on baseline arterial blood pressure, suggesting that this peptide does not play an important role in modulating this parameter under physiological conditions. However, these anesthetized rats had an intact CNS and, hence, operative baroreceptor reflex mechanisms could have masked reflex sympathetic hyperactivity (see below). At this point, we could imply that acute disruption of the CGRPergic system by gepants has no substantial effects on cardiovascular homeostasis, but a chronic CGRP disruption by antibodies against CGRP or its receptor (compounds with a long half-life) could unmask vasopressor mechanisms and, thus, the physiological relevance of CGRP. In this respect, initial Phase 1 RCT using fremanezumab (TEV-48125, formerly LBR-101 or RN307), a humanized antibody that binds both isoforms of CGRP, showed that i.v. administration of this antibody (0.2–2000 mg) not only is well tolerated [140], but also has no clinical cardiovascular or hemodynamic effects; these results were observed after a single i.v. administration and were followed for 168 days [134]. Furthermore, Walter et al. [141] reported a lack of systemic cardiovascular alterations after single (100 mg/kg; i.v.) or multiple-dose (10–300 mg/kg; i.v.; once weekly for 14 weeks) of this compound in monkeys.

Nevertheless, some aspects of the hemodynamic actions of CGRP receptor antagonists can be detected only in the absence of central baroreflex mechanisms. Indeed, using a pithed rat model (in which the CNS is not operative), Avilés-Rosas et al. [142] have shown that single i.v. injections of olcegepant blocked both the neurogenic (electrically induced) and non-neurogenic (i.v. α-CGRP administration) CGRPergic vasodepressor responses; this acute blockade by olcegepant potentiated the neurogenic and non-neurogenic noradrenergic vasopressor responses, an effect that might result in an increased vascular resistance and, consequently, in a prohypertensive action [142]. This olcegepant's prohypertensive action may be even more pronounced (though not yet proven) when using monoclonal antibodies against CGRP or its receptor. Evidently, this possibility requires further research.

Most importantly, coronary arteries are innervated by a dense supply of CGRP-positive fibers [143], and in human isolated coronary arteries olcegepant [119] and telcagepant [120] antagonized the CGRP-induced vasodilation. If this blockade occurred chronically (e.g. when using CGRP (receptor) antibodies), it may result in cardiovascular disease and, perhaps, in facilitation of transient heart or brain ischemia leading to a cardiac or brain infarction [144].

Indeed, in CGRP knockout mice, where blood pressure was continuously recorded using telemetry, blood pressure was increased, an effect partly due to an increase in the sympathetic discharge [145]. These findings: (i) support a previous report suggesting that CGRPergic nerves modulate sympathetic activity [146] and (ii) contrast with a previous report using knockout mice where no change in blood pressure was observed [147]. Nevertheless, in this latter case, blood pressure was measured at one time in a day instead a continuous recording of blood pressure.

Although exogenous CGRP administration results in a decrease in blood pressure in humans [148], acute treatment with olcegepant does not have a significant effect in this hemodynamic variable [130,149]. Certainly, acute CGRP system disruption by gepants cannot be extrapolated with chronic effects induced by the monoclonal antibodies against CGRP or its receptor. The cardiovascular effects related with a long-term blockade of the CGRP system and multiple-dose are currently unknown; thus, studies on this issue in large scale clinical use (i.e. Phase IV) could shed further light on this topic.

3.5. Implications for future research

Although migraine treatment with nonspecific drugs (e.g. some NSAIDs) is useful to treat mild-to-moderate migraine attacks, the search for new molecules has been fostered considering that classical specific antimigraine drugs (i.e. ergots and triptans) exert critical side effects. Indeed, the antimigraine drugs currently available (apart from limited <50% efficacy) produce a high number of adverse side effects leading to poor compliance and adherence [150]. In this context, the advent of gepants and, most recently, of the antibodies against CGRP or its receptor, emphasizes the relevance of CGRP in migraine pathophysiology [151]. However, migraine is a more complex disorder and CGRP – though relevant – is not the only molecule involved. Certainly, other pain-producing molecules could contribute to the development of headache in migraine attacks, and blocking only CGRP transmission would not be enough to achieve an optimal efficacy.

The use of antibodies against CGRP or its receptor for migraine therapy represents a new hope for this complex disorder, but some interesting studies show that in all Phase II RCT a high placebo response rate was also observed [see Table 3 for references]; these results could mask the net antimigraine effect of these antibodies. Clearly, chronic blockade of CGRPergic function as a prophylactic antimigraine strategy could raise potential harmful consequences, as previously pointed out [13]. An important recommendation for medical practitioners is to evaluate the pros and cons involved in selecting CGRP blockade [151].

Exploration of other pathways involved in migraine pathophysiology remains as an active option to find novel and more effective treatments. Among those potential alternatives for treating migraine (with aura), molecules that interfere with neuronal-glia communication via interfering with gap junction complexes (e.g. tonabersat) have been considered [152]. Indeed, in animal models of migraine, tonabersat inhibits CSD, neurogenic inflammation, and carotid vasodilation induced by trigeminal ganglion stimulation without relevant effects in the cardiovascular system and CNS [153–155]. However, an RCT (Phase II trial) showed that tonabersat did not have a better effect than placebo [156]. Certainly, this molecule was well tolerated and no major adverse events were reported [156].

3.6. Further pharmacological considerations

An issue that complicates migraine treatment is comorbidities (e.g. cardiovascular pathologies, psychiatric diseases, epilepsy, asthma, etc.) [157,158]. These comorbidities increase the possibility for drug–drug interactions, augment potential side effects, and limit the safest treatment choice [159]. Another clinical consideration that must be analyzed here is the effect resulting from the combination of prophylactic and acute medications. Since classic antimigraine drugs (e.g. ergots, triptans, etc.) are associated with vascular actions [160], the combination of two or more of these compounds may be especially dangerous in migraineurs with (a propensity to) cardiovascular pathologies. For example, ergot-based compounds should not be combined with triptans (e.g. eletriptan) [161] as the resulting vasocontractile effects may be too intense. In this case, ditans such as lasmiditan may represent a better therapeutic choice as it is completely devoid of vasoconstrictor actions [162] if tolerability studies confirm its safety profile [53,54].

4. Expert opinion

Much effort has been put on the development of antimigraine pharmacotherapies and the search for new compounds with antimigraine properties has been fostered considering that apart from its low efficacy in the population (<50%), the adverse side effects of current antimigraine pharmacotherapy remain as a main therapeutic issue [4,9,12]. Certainly, the triptans (5-$HT_{1B/1D}$ receptor agonists) are relatively safe and well tolerated if used in correct dosage and frequency, but MOH is one of the most serious complications linked to triptans overuse. In addition to the nonresponder patients, subjects with cardiovascular risks are not suitable for the use of these compounds, as they have the propensity to produce coronary vasoconstriction [49].

The fact that some triptans can also activate 5-HT$_{1F}$ receptors, and that this activation does not result in vasoconstriction, led to the development of the ditans (in this review defined as selective 5-HT$_{1F}$ receptor agonists [9,12]). This culminated in the demonstration that the lipophilic 5-HT$_{1F}$ receptor agonist, lasmiditan, was effective in the acute treatment of migraine [53,54], but it also produced some CNS adverse effects that should be taken into account in clinical practice. Nevertheless, being devoid of vasoconstrictor effects in human blood vessels [84], lasmiditan has a cardiovascular safety advantage over the triptans.

The subsequent advent of antibodies against CGRP or its receptor targeting the CGRPergic transmission may seem to be the 'magic bullet' against migraine (episodic and chronic), but this disorder is complex. The fact that not only CGRP may be responsible for the development of migraine attacks is evident from recent data showing that, although they show a consistent response, the antibodies against CGRP or its receptor do not provide a complete relief in all patients (for review, see [151]). The future challenge will be to identify which patients will qualify for treatment with antibodies against CGRP or its receptor, because they will benefit from such a treatment, preferably by a biomarker that can be assessed prior to treatment. Alternatively, patients where other, non-CGRPergic mechanisms may play a more prominent role should also be identified so that they can be treated with other (non-CGRPergic) medications. Although the monoclonal antibodies against CGRP or its receptor show excellent short-term tolerability, further points concerning long-term safety need to be clarified. Moreover, since the mechanisms of action of these monoclonal antibodies differ from those of the triptans, it is important that their effects could be assessed in adequate models. Certainly, new treatments come with new challenges and, as noted by Paracelsus 'All things are poison and nothing is without poison, only the dose makes a thing not a poison'. Evidently, CGRP receptor antagonists and monoclonal CGRP (receptor) antibodies have emerged as key compounds against migraine, but the further development of some of the small molecule antagonists was suspended because they produced hepatotoxicity, whereas the latter are apparently devoid of important adverse side effects (as compared to placebo). Clearly, long term (i.e. Phase III and IV) studies are required to definitely delineate the adverse side effect potential and safety of the antibodies against CGRP or its receptor.

Funding

This work was partially supported by the following research grants: (i) Programa de Apoyo a Proyectos de Investigación e Innovación Tecnológica (PAPIIT-UNAM-Mexico, grant No. IA203117 for Abimael González-Hernández); and (ii) Consejo Nacional de Ciencia y Tecnología (CONACyT-Mexico, Grant No. 219707 for Carlos M. Villalón)

Declaration of interest

A MaassenVanDenBrink has received grants from Amgen, Novartis, Lilly and CoLucid. The authors have no other relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript apart from those disclosed. Peer reviewers on this manuscript have no relevant financial or other relationships to disclose

ORCID

Abimael González-Hernández http://orcid.org/0000-0002-6472-1419

References

Papers of special note have been highlighted as either of interest (•) or of considerable interest (••) to readers.

1. Olesen J, Burstein R, Ashina M, et al. Origin of pain in migraine: evidence for peripheral sensitization. Lancet Neurol. 2009;8:679–690.
2. Steiner TJ, Stovner J, Birbeck GL. Migraine: the seventh disabler. Cephalalgia. 2013;33:289–290.
3. Andlin-Sobocki P, Jonsson B, Wittchen HU, et al. Cost of disorders of the brain in Europe. Eur J Neurol. 2005;12(Suppl 1):1–27.
4. Villalón CM, Olesen J. The role of CGRP in the pathophysiology of migraine and efficacy of CGRP receptor antagonists as acute antimigraine drugs. Pharmacol Ther. 2009;124:309–323.
5. Edvinsson L, Villalón CM, MaassenVanDenBrink A. Basic mechanisms of migraine and its acute treatment. Pharmacol Ther. 2012;136:319–333.
6. Goadsby PJ, Holland PR, Martins-Oliveira M, et al. Pathophysiology of migraine: a disorder of sensory processing. Physiol Rev. 2017;97:553–622.
7. Akerman S, Holland PR, Goadsby PJ. Diencephalic and brainstem mechanisms in migraine. Nat Rev Neurosci. 2011;12:570–584.
8. Villalón CM, MaassenVanDenBrink A. The role of 5-hydroxytryptamine in the pathophysiology of migraine and its relevance to the design of novel treatments. Mini Rev Med Chem. 2017;17:928–938.
9. Villalón CM, Centurión D, Valdivia LF, et al. Migraine: pathophysiology, pharmacology, treatment and future trends. Curr Vasc Pharmacol. 2003;1:71–84.
10. Olesen J, Goadsby PJ, Ramadan N, et al. The headaches. In: Olesen J, Goadsby PJ, Ramadan N, et al. editors. The Headaches. 3rd ed. Philadelphia: Lippincott Williams & Wilkins; 2006. p. 1–1169.
11. Humphrey PP. The discovery of a new drug class for the acute treatment of migraine. Headache. 2007;47(Suppl 1):S10–S19.
12. Chan KY, Vermeersch S, de Hoon J, et al. Potential mechanisms of prospective antimigraine drugs: a focus on vascular (side) effects. Pharmacol Ther. 2011;129:332–351.
13. MaassenVanDenBrink A, Meijer J, Villalón CM, et al. Wiping out CGRP: potential cardiovascular risks. Trends Pharmacol Sci. 2016;37:779–788.
14. Pellesi L, Guerzoni S, Pini LA. Spotlight on anti-CGRP monoclonal antibodies in migraine: the clinical evidence to date. Clin Pharmacol Drug Dev. 2017. DOI:10.1002/cpdd.345
15. Dib M. Optimizing prophylactic treatment of migraine: subtypes and patient matching. Ther Clin Risk Management. 2008;4:1061–1078.
16. Dahlöf C, MaassenVanDenBrink A. Dihydroergotamine, ergotamine, methysergide and sumatriptan - basic science in relation to migraine treatment. Headache. 2012;52:707–714.
17. Ramírez Rosas MB, Labruijere S, Villalón CM, et al. Activation of 5-hydroxytryptamine$_{1B/1D/1F}$ receptors as a mechanism of action of antimigraine drugs. Expert Opin Pharmacother. 2013;14:1599–1610.
18. Gross RA, Johnston KC. Levels of evidence taking Neurology® to the next level. Neurology. 2009;72:8–10.
19. Evers S, Áfra J, Frese A, et al. EFNS guideline on the drug treatment of migraine – revised report of an EFNS task force. Eur J Neurol. 2009;16:968–981.

20. Fumal A, Schoenen J. Current migraine management – patient acceptability and future approaches. Neuropsych Dis Treat. 2008;4:1043–1057.
21. Bolay H, Durham P. Pharmacology. In: Nappi G, Moskowitz M, editors. Headache. The Netherlands: Elsevier; 2011. p. 47–71.
22. Brack A, Rittner HL, Schäfer M. Nichtopioidanalgetika zur perioperativen Schmerztherapie [Non-opioid analgesics for perioperative pain therapy. Risks and rational basis for use]. Anaesthesist. 2004;53: 263–280. German.
23. Marmura MJ, Silberstein SD, Schwedt TJ. The acute treatment of migraine in adults: the American Headache Society evidence assessment of migraine pharmacotherapies. Headache. 2015;55:3–20.
24. Silberstein SD, Holland S, Freitag F, et al. Evidence-based guideline update: pharmacologic treatment for episodic migraine prevention in adults: report of the Quality Standards Subcommittee of the American Academy of Neurology and the American Headache Society. Neurology. 2012;78:1337–1345.
25. Holland S, Silberstein SD, Freitag F, et al. Evidence-based guideline update: NSAIDs and other complementary treatments for episodic migraine prevention in adults: report of the quality standards subcommittee of the American Academy of Neurology and the American Headache Society. Neurology. 2012;78:1345.
26. Goadsby PJ, Liptom RB, Ferrari MD. Migraine – current understanding and treatment. N Engl J Med. 2002;346:257–270.
27. Dodick D, Lipton RB, Martin V, et al. Consensus statement: cardiovascular safety profile of triptans (5-HT agonists) in the acute treatment of migraine. Headache. 2004;44:414–425.
28. Alexander SPH, Davenport AP, Kelly E, et al. The concise guide to PHARMACOLOGY 2015/16: G protein-coupled receptors. Br J Pharmacol. 2015;172:5744–5869.
29. Law V, Knox C, Djoumbou Y, et al. DrugBank 4.0: shedding new light on drug metabolism. Nucleic Acids Res. 2014;42:D1091–D1097.
30. Dodick DW, Goadsby PJ, Spierings EL, et al. Safety and efficacy of LY2951742, a monoclonal antibody to calcitonin gene-related peptide, for the prevention of migraine: a phase 2, randomised, double-blind, placebo-controlled study. Lancet Neurol. 2014;13:885–892.
31. Dodick DW, Goadsby PJ, Silberstein SD, et al. Safety and efficacy of ALD403, an antibody to calcitonin gene-related peptide, for the prevention of frequent episodic migraine: a randomised, double-blind, placebo-controlled, exploratory phase 2 trial. Lancet Neurol. 2014;13:1100–1107.
32. Bigal ME, Dodick DW, Rapoport AM, et al. Safety, tolerability, and efficacy of TEV-48125 for preventive treatment of high-frequency episodic migraine: a multicentre, randomised, double-blind, placebo-controlled, phase 2b study. Lancet Neurol. 2015;14:1081–1090.
33. Bigal ME, Edvinsson L, Rapoport AM, et al. Safety, tolerability, and efficacy of TEV-48125 for preventive treatment of chronic migraine: a multicentre, randomised, double-blind, placebo-controlled, phase 2b study. Lancet Neurol. 2015;14:1091–1100.
34. Bigal ME, Dodick DW, Krymchantowski AV, et al. TEV-48125 for the preventive treatment of chronic migraine. Efficacy at early time points. Neurology. 2016;87:41–48.
35. Benemei S, Nicoletti P, Capone JG, et al. CGRP receptors in the control of pain and inflammation. Curr Opin Pharmacol. 2009;9:9–14.
36. Hou M, Xing H, Cai Y, et al. The effect and safety of monoclonal antibodies to calcitonin gene-related peptide and its receptor on migraine: a systematic review and meta-analysis. J Headache Pain. 2017;18:42.
37. Cashman JN. The mechanisms of action of NSAIDs in analgesia. Drugs. 1996;52(S5):13–23.
38. Aydin D, Vuralli D, Akçali DT, et al. Metoclopramide inhibits trigeminovascular activation: evidence for effective acute attack treatment in migraine. Turk J Med Sci. 2017;47:343–347.
39. Mitsikostas DD, Reuter U. Calcitonin gene-related peptide monoclonal antibodies for migraine prevention: comparisons across randomized controlled studies. Curr Opin Neurol. 2017;30:272–280.
40. Silberstein SD, McCrory DC. Ergotamine and dihydroergotamine: history, pharmacology, and efficacy. Headache. 2003;43:144–166.
41. Kilinc E, Guerrero-Toro C, Zakharov A, et al. Serotonergic mechanisms of trigeminal meningeal nociception: implications for migraine pain. Neuropharmacology. 2017;116:160–173.
42. Valdivia LF, Centurión D, Arulmani U, et al. 5-HT_{1B} receptors, $alpha_{2A/2C}$- and, to a lesser extent, $alpha_1$-adrenoceptors mediate the external carotid vasoconstriction to ergotamine in vagosympathectomised dogs. Naunyn-Schmiedeberg's Arch Pharmacol. 2004;370:46–53.
43. Marichal-Cancino BA, González-Hernández A, Manrique-Maldonado G, et al. Intrathecal dihydroergotamine inhibits capsaicin-induced vasodilatation in the canine external carotid circulation via GR127935- and rauwolscine-sensitive receptors. Eur J Pharmacol. 2012;692:69–77.
44. Villalón CM, Galicia-Carreón J, González-Hernández A, et al. Pharmacological evidence that spinal $\alpha_{(2C)}$- and, to a lesser extent, $\alpha_{(2A)}$-adrenoceptors inhibit capsaicin-induced vasodilatation in the canine external carotid circulation. Eur J Pharmacol. 2012;683:204–210.
45. Pradalier A, Lantéri-Minet M, Géraud G, et al. The PROMISE study: PROphylaxis of MIgraine with SEglor (dihydroergotamine mesilate) in French primary care. CNS Drugs. 2004;18:1149–1163.
46. Akin D, Gurdal H. Involvement of 5-HT_{1B} and 5-HT_{1D} receptors in sumatriptan mediated vasocontractile response in rabbit common carotid artery. Br J Pharmacol. 2002;136:177–182.
47. Jeong HJ, Mitchell VA, Vaughan CW. Role of 5-$HT_{(1)}$ receptor subtypes in the modulation of pain and synaptic transmission in rat spinal superficial dorsal horn. Br J Pharmacol. 2012;165:1956–1965.
48. Wackenfors A, Jarvius M, Ingemansson R, et al. Triptans induce vasoconstriction of human arteries and veins from the thoracic wall. J Cardiovasc Pharmacol. 2005;45:476–484.
49. MaassenVanDenBrink A, Reekers M, Bax WA, et al. Coronary side-effect potential of current and prospective antimigraine drugs. Circulation. 1998;98:25–30.
50. Moschiano F, Allais G, Grazzi L, et al. Naratriptan in the short-term prophylaxis of pure menstrual migraine. Neurol Sci. 2005;26(Suppl 2):162–166.
51. Hu Y, Guan X, Fan L, et al. Triptans in prevention of menstrual migraine: a systematic review with meta-analysis. J Head Pain. 2013;14:7.
52. Christensen AF, Esserlind AL, Werge T, et al. The influence of genetic constitution on migraine drug responses. Cephalalgia. 2016;36:624–639.
53. Färkkilä M, Diener HC, Géraud G, et al. Efficacy and tolerability of lasmiditan, an oral 5-HT_{1F} receptor agonist, for the acute treatment of migraine: a phase 2 randomised, placebo-controlled, parallel-group, dose-ranging study. Lancet Neurol. 2012;11:405–413.
54. Ferrari MD, Färkkilä M, Reuter U, et al. Acute treatment of migraine with the selective 5-HT_{1F} receptor agonist lasmiditan - a randomised proof-of-concept trial. Cephalalgia. 2010;30:1170–1178.
55. Tfelt-Hansen P, Olesen J. Possible site of action of CGRP antagonists in migraine. Cephalalgia. 2011;31:748–750.
56. Doods H, Hallermayer G, Wu D, et al. Pharmacological profile of BIBN4096BS, the first selective small molecule CGRP antagonist. Br J Pharmacol. 2000;129:420–423.
57. Eftekhari S, Salvatore CA, Johansson S, et al. Localization of CGRP, CGRP receptor, PACAP and glutamate in trigeminal ganglion relation to the blood–brain barrier. Brain Res. 2015;1600:93–109.
58. Walker CS, Eftekhari S, Bower RL, et al. A second trigeminal CGRP receptor: function and expression of the AMY1 receptor. Ann Clin Transl Neurol. 2015;2:595–608.
59. Wu H, Dall'Acqua WF. Humanized antibodies and their applications. Methods. 2005;36:1–2.
60. Goldstein J, Hagen M, Gold M. Results of a multicenter, double-blind, randomized, parallel-group, placebo-controlled, single-dose study comparing the fixed combination of acetaminophen, acetylsalicylic acid, and caffeine with ibuprofen for acute treatment of patients with severe migraine. Cephalalgia. 2014;34:1070–1078.

61. Tajti J, Majláth Z, Szok D, et al. Drug safety in acute migraine treatment. Exp Op Drug Saf. 2015;14:891–909.
62. Linde K, Rossnagel K. Propranolol for migraine prophylaxis. Cochrane Database Syst Rev. 2004;2:CD003225.
63. Boyer N, Signoret-Genest J, Artola A, et al. Propranolol treatment prevents chronic central sensitization induced by repeated dural stimulation. Pain. 2017;158:2025–2034.
64. Shields KG, Goadsby PJ. Propranolol modulates trigeminovascular responses in thalamic ventroposteromedial nucleus: a role in migraine? Brain. 2004;128:86–97.
65. Koella WP. CNS-related (side-) effects of β-blockers with special reference to mechanisms of action. Eur J Clin Pharmacol. 1985;28:55–63.
66. Tekes K, Szegi P, Hashemi F, et al. Medicinal chemistry of antimigraine drugs. Curr Med Chem. 2013;20:3300–3316.
67. Wang DW, Mistry AM, Kahlig KM, et al. Propranolol blocks cardiac and neuronal voltage-gated sodium channels. Front Pharmacol. 2010;1:144.
68. Diener HC, Bussone G, Van Oene JC, et al. Topiramate reduces headache days in chronic migraine: a randomized, double-blind, placebo-controlled study. Cephalalgia. 2007;27:814–823.
69. Lakshmi CV, Singhi P, Malhi P, et al. Topiramate in the prophylaxis of pediatric migraine: a double-blind placebo-controlled trial. J Child Neurol. 2007;22:829–835.
70. Shank RP, Gardocki JF, Streeter AJ, et al. An overview of the preclinical aspects of topiramate: pharmacology, pharmacokinetics, and mechanism of action. Epilepsia. 2000;41(Suppl 1):3–9.
71. Mathew NT, Kailasam J, Meadors L, et al. Intravenous valproate sodium (depacon) aborts migraine rapidly: a preliminary report. Headache. 2000;40:720–723.
72. Hering R, Kuritzky A. Sodium valproate in the prophylactic treatment of migraine: a double-blind study versus placebo. Cephalalgia. 1992;12:81–84.
73. Grosser T, Smyth E, FitzGerald GA. Anti-inflammatory, antipyretic, and analgesic agents: pharmacotherapy of gout. In: Brunton LL, editor. Goodman & Gilman's the pharmacological basis of therapeutics. 12th ed. USA: McGraw-Hill; 2011. p. 954–1004.
74. Tacconelli S, Bruno A, Grande R, et al. Nonsteroidal anti-inflammatory drugs and cardiovascular safety-translating pharmacological data into clinical readouts. Exp Op Drug Saf. 2017;16:791–807.
75. Meek IL, van de Laar MAFJ, Vonkeman HE. Non-steroidal anti-inflammatory drugs: an overview of cardiovascular risks. Pharmaceuticals. 2010;3:2146–2162.
76. Havanka H, Dahlöf C, Pop PH, et al. Efficacy of naratriptan tablets in the acute treatment of migraine: a dose-ranging study. Clin Ther. 2000;22:970–980.
77. Pfaffenrath V, Cunin G, Sjonell G, et al. Efficacy and safety of sumatriptan tablets (25 mg, 50 mg, and 100 mg) in the acute treatment of migraine: defining the optimum doses of oral sumatriptan. Headache. 1998;38:184–190.
78. Rapoport AM, Ramadan NM, Adelman JU, et al. Optimizing the dose of zolmitriptan (Zomig,* 311C90) for the acute treatment of migraine A multicenter, double-blind, placebo-controlled, dose range-finding study. Neurology. 1997;49:1210–1218.
79. Meyler WJ. Side effects of ergotamine. Cephalalgia. 1996;16:5–10.
80. Tfelt-Hansen P, Saxena PR, Ferrari MD. Ergot alkaloids. In: de Wolff FA, editor. Handbook of clinical neurology, Vol. 21 (65): intoxications of the nervous system, part II. Vol. 21. Amsterdam: Elsevier Science BV; 1995. p. 61–78.
81. Andersen AR, Tfelt-Hansen P, Lassen NA. The effect of ergotamine and dihydroergotamine on cerebral blood flow in man. Stroke. 1987;18:120–123.
82. Flexman AM, Del Vicario G, Schwarz S. Dark green blood in the operating theatre. Lancet. 2007;369:1972.
83. Fard JK, Hamzeiy H, Sattari M, et al. Triazole rizatriptan induces liver toxicity through lysosomal/mitochondrial dysfunction. Drug Res. 2016;66:470–478.
84. Lucaites VL, Krushinski JH, Schaus JM, et al. [3H] LY334370, a novel radioligand for the 5-HT$_{1F}$ receptor. II. Autoradiographic localization in rat, guinea pig, monkey and human brain. Naunyn Schmiedeberg´S Arch Pharmacol. 2005;371:178–184.
85. Varnäs K, Hurd YL, Hall H. Regional expression of 5-HT$_{1B}$ receptor mRNA in the human brain. Synapse. 2005;56:21–28.
86. O'Quinn S, Davis RL, Guttermann DL, et al. Prospective large-scale study of the tolerability of subcutaneous sumatriptan injection for the acute treatment of migraine. Cephalalgia. 1999;19:223–231.
87. Welch KMA, Mathew NT, Stone P, et al. Tolerability of sumatriptan: clinical trials and post-marketing experience. Cephalalgia. 2000;20:687–695.
88. Rubio-Beltran E, Kristian H, Labastida A, et al. Lasmiditan and sumatriptan: comparison of in vivo vascular constriction in the dog and in vitro contraction of human arteries. Cephalalgia. 2016;36(1S):104–105.
89. Edvinsson L, Tfelt-Hansen P. The blood brain barrier in migraine treatment. Cephalalgia. 2008;28:1245–1258.
90. Tfelt-Hansen PC. Does sumatriptan cross the blood-brain barrier in animals and man. J Headache Pain. 2010;11:5–12.
91. Hoskin KL, Goadsby PJ. Comparison of more and less lipophilic serotonin (5-HT$_{1B/1D}$) agonists in a model of trigeminovascular nociception in cat. Exp Neurol. 1998;150:45–51.
92. Jiménez-Mena LR, Gupta S, Muñoz-Islas E, et al. Clonidine inhibits the canine external carotid vasodilatation to capsaicin by a adrenoceptors. Eur J Pharmacol. 2006;543:68–76.
93. Kaube H, Hoskin KL, Goadsby PJ. Inhibition by sumatriptan of central trigeminal neurones only after blood–brain barrier disruption. Br J Pharmacol. 1993;109:788–792.
94. Muñoz-Islas E, Lozano-Cuenca J, González-Hernández A, et al. Spinal sumatriptan inhibits capsaicin-induced canine external carotid vasodilatation via 5-HT$_{1B}$ rather than 5-HT$_{1D}$ receptors. Eur J Pharmacol. 2009;615:133–138.
95. Weiller C, May A, Limroth V, et al. Brain stem activation in spontaneous human migraine attacks. Nat Med. 1995;1:658–660.
96. Schankin CJ, Maniyar FH, Seo Y, et al. Ictal lack of binding t brain parenchyma suggest integrity of the blood-brain barrier for 11C-dihydroergotamine during glyceryl trinitrate-induced migraine. Brain. 2016;139:1994–2001.
97. Ala-Hurula V, Myllyla VV, Arvela P, et al. Systemic availability of ergotamine tartrate after three successive doses and during continuous medication. Eur J Clin Pharmacol. 1979;16:355–360.
98. Hougaard A, Amin FM, Christensen CE, et al. Increased brainstem perfusion, but no blood-brain barrier disruption, during attacks of migraine with aura. Brain. 2017;2017(140):1633–1642.
99. Villalón CM, De Vries P, Rabelo G, et al. Canine external carotid vasoconstriction to methysergide, ergotamine and dihydroergotamine: role of 5-HT$_{1B/1D}$ receptors and alpha$_2$-adrenoceptors. Br J Pharmacol. 1999;126:585–594.
100. Cortijo J, Marti-Cabrera M, Bernabeu E, et al. Characterization of 5-HT receptors on human pulmonary artery and vein: functional and binding studies. Br J Pharmacol. 1997;122:1455–1463.
101. Labruijere S, Chan KY, de Vries R, et al. Dihydroergotamine and sumatriptan in isolated human coronary artery, middle meningeal artery and saphenous vein. Cephalalgia. 2015;35:182–189.
102. Chan KY, De Vries R, Leijten FP, et al. Functional characterization of contractions to tegaserod in human isolated proximal and distal coronary arteries. Eur J Pharmacol. 2009;619:61–67.
103. De Hoon JN, Willigers JM, Troost J, et al. Vascular effects of 5-HT$_{1B/1D}$-receptor agonists in patients with migraine headaches. Clin Pharmacol Ther. 2000;68:418–426.
104. Vanmolkot FH, De Hoon JN. Acute effects of sumatriptan on aortic blood pressure, stiffness, and pressure waveform. Clin Pharmacol Ther. 2006;80:85–94.
105. Fishman GI, Chugh S, DiMarco JP, et al. Sudden cardiac death prediction and prevention report from a National Heart, Lung, and Blood Institute and Heart Rhythm Society workshop. Circulation. 2010;122:2335–2348.
106. Diener HC, Holle D, Solbach K, et al. Medication-overuse headache: risk, factors, pathophysiology and management. Nat Rev Neurol. 2016;12:575–583.

107. Chiang -C-C, Schwedt TJ, Wang S-J, et al. Treatment of medication-overuse headache: a systematic review. Cephalalgia. 2016;36:371–386.
108. Westergaard ML, Hansen EH, Glumer C, et al. Definitions of medication overuse headache in population-based studies and their implications on prevalence estimates: a systematic review. Cephalalgia. 2014;34:409–425.
109. Westergaard ML, Glumer C, Hansen EH, et al. H. Prevalence of chronic headache with and without medication overuse: associations with socioeconomic position and physical and mental health status. Pain. 2014;155:2005–2013.
110. Evers S, Marziniak M. Clinical features, pathophysiology, and treatment of medication-overuse headache. Lancet Neurol. 2010;9:391–401.
111. Limmroth V, Kazarawa S, Fritsche G, et al. Features of medication overuse headache following overuse of different acute headache drugs. Neurology. 2002;59:1011–1014.
112. Calabresi P, Cupini LM. Medication-overuse headache: similarities with drug addiction. Trends Pharmacol Sci. 2005;26:62–68.
113. De Felice M, Ossipov MH, Wang R, et al. Triptan-induced latent sensitization: a possible basis for medication overuse headache. Ann Neurol. 2010;67:325–337.
114. De Felice M, Ossipov MH, Wang R, et al. Triptan-induced enhancement of neuronal nitric oxide synthase in trigeminal ganglion dural afferents underlies increased responsiveness to potential migraine triggers. Brain. 2010;133:2475–2488.
115. Supornsilpchai W, Le Grand SM, Srikiatkhachorn A. Cortical hyperexcitability and mechanism of medication-overuse headache. Cephalalgia. 2010;30:1101–1109.
116. Schwedt TJ, Chong CD. Medication overuse headache: pathophysiological insights from structural and functional brain MRI research. Headache. 2017;57:1173–1178.
117. Bongsebandhu-Phubhakdi S, Srikiatkhachorn A. Pathophysiology of medication-overuse headache: implications from animal studies. Curr Pain Headache Rep. 2012;16:110–115.
118. Hyman SE, Malenka RC. Addiction and the brain: the neurobiology of compulsion and its persistence. Nat Rev Neurosci. 2001;2:695–703.
119. Gupta S, Mehrotra S, Villalón CM, et al. Characterisation of CGRP receptors in human and porcine isolated coronary arteries: evidence for CGRP receptor heterogeneity. Eur J Pharmacol. 2006;530:107–116.
120. Chan KY, Edvinsson L, Eftekhari S, et al. Characterization of the CGRP receptor antagonist telcagepant (MK-0974) in human isolated coronary arteries. J Pharmacol Exp Ther. 2010;334:746–752.
121. Behm MO, Blanchard RL, Murphy MG, et al. Assessment of the effect of MK-0974, an oral CGRP receptor antagonist, on spontaneous ischemia in patients with stable cardiovascular disease. Headache. 2008;48(Suppl 1):S39.
122. Russell FA, King R, Smillie SJ, et al. Calcitonin gene-related peptide: physiology and pathophysiology. Physiol Rev. 2014;94:459–466.
123. Iovino M, Feifel U, Yong CL, et al. Safety, tolerability and pharmacokinetics of BIBN 4096 BS, the first selective small molecule calcitonin gene-related peptide receptor antagonist, following single intravenous administration in healthy volunteers. Cephalalgia. 2004;24:645–656.
124. Olesen J, Diener HC, Husstedt IW, et al. Calcitonin gene-related peptide receptor antagonist BIBN 4096BS for the acute treatment of migraine. N Engl J Med. 2004;350:1104–1110.
125. Ho TW, Mannix LK, Fan X, et al. Randomized controlled trial of an oral CGRP receptor antagonist, MK-0974, in acute treatment of migraine. Neurology. 2008;70:1304–1312.
126. Ho TW, Ferrari MD, Dodick DW, et al. Efficacy and tolerability of MK-0974 (telcagepant), a new oral antagonist of calcitonin gene-related peptide receptor, compared with zolmitriptan for acute migraine: a randomised, placebo-controlled, parallel-treatment trial. Lancet. 2008;372:2115–2123.
127. Connor KM, Shapiro RE, Diener HC, et al. Randomized, controlled trial of telcagepant for the acute treatment of migraine. Neurology. 2009;73:970–977.
128. Silberstein SD. Emerging target-based paradigms to prevent and treat migraine. Clin Pharmacol Ther. 2013;93:78–85.
129. Hewitt DJ, Aurora SK, Dodick DW, et al. Randomized controlled trial of the CGRP receptor antagonist MK-3207 in the acute treatment of migraine. Cephalalgia. 2011;31:712–722.
130. Diener HC, Barbanti P, Dahlöf C, et al. BI 44370 TA, an oral CGRP antagonist for the treatment of acute migraine attacks: results from a phase II study. Cephalalgia. 2011;31:573–584.
131. Marcus R, Goadsby PJ, Dodick D, et al. BMS-927711 for the acute treatment of migraine: a double-blind, randomized, placebo controlled, dose-ranging trial. Cephalalgia. 2014;34:114–125.
132. Pardridge WM, Eisenberg J, Yang J. Human blood–brain barrier insulin receptor. J Neurochem. 1985;44:1771–1778.
133. Wang Y, Welty DR. The simultaneous estimation of the influx and efflux blood-brain barrier permeabilities of gabapentin using a microdialysis-pharmacokinetic approach. Pharm Res. 1996;13:398–403.
134. Bigal ME, Walter S, Bronson M, et al. Cardiovascular and hemodynamic parameters in women following prolonged CGRP inhibition using LBR-101, a monoclonal antibody against CGRP. Cephalagia. 2014;34:968–976.
135. Novartis announces Phase III study shows AMG 334 significantly reduces monthly migraine days in people with episodic migraine [Internet]. [cited Oct 2017]. Available from: https://www.novartis.com/news/media-releases/novartis-announces-phase-iii-study-shows-amg-334-significantly-reduces-monthly
136. Sun H, Dodick DW, Silberstein S, et al. Safety and efficacy of AMG 334 for prevention of episodic migraine: a randomised, double-blind, placebo-controlled, phase 2 trial. Lancet Neurol. 2016;15:382–390.
137. Taguchi T, Kawasaki H, Imamura T, et al. Endogenous calcitonin gene-related peptide mediates nonadrenergic noncholinergic depressor response to spinal cord stimulation in the pithed rat. Circ Res. 1992;71:357–364.
138. González-Hernández A, Marichal-Cancino BA, Lozano-Cuenca J, et al. Heteroreceptors modulating CGRP release at neurovascular junction: potential therapeutic implications on some vascular-related diseases. BioMed Res Int. 2016;2016:2056786.
139. Arulmani U, Schuijt MP, Heiligers JPC, et al. Effects of the calcitonin gene-related peptide (CGRP) receptor antagonist BIBN4096BS on α-CGRP induced regional haemodynamic changes in anaesthetised rats. Basic Clin Pharmacol Toxicol. 2004;94:291–297.
140. Bigal ME, Escandon R, Bronson M, et al. Safety and tolerability of LBR-101, a humanized monoclonal antibody that blocks the binding of CGRP to its receptor: results of the phase 1 program. Cephalalgia. 2014;34:483–492.
141. Walter S, Alibhoy A, Escandon R. Evaluation of cardiovascular parameters in cynomolgus monkeys following IV administration of LBR-101, a monoclonal antibody against calcitonin gene-related peptide. MAbs. 2014;6:871–878.
142. Avilés-Rosas VH, Rivera-Mancilla E, Marichal-Cancino BA, et al. Olcegepant blocks neurogenic and non-neurogenic CGRPergic vasodepressor responses and facilitates noradrenergic vasopressor responses in pithed rats. Br J Pharmacol. 2017;174:2001–2014.
143. Gulbenkian S, Opgaard OS, Ekman R, et al. Peptidergic innervation of human epicardial coronary arteries. Circ Res. 1993;73:579–588.
144. Pascual J. CGRP antibodies: the Holy Grail for migraine prevention? Lancet Neurol. 2015;14:1066–1067.
145. Mai TH, Wu J, Diedrich A, et al. Calcitonin gene- peptide (CGRP) in autonomic cardiovascular regulation and vascular structure. J Am Soc Hyper. 2014;8:286–296.
146. Oh-Hashi Y, Shindo T, Kurihara Y, et al. Elevated sympathetic nervous activity in mice deficient in α-CGRP. Circulation. 2001;89:983–990.
147. Smillie SJ, King R, Kodji X, et al. An ongoing role of α-calcitonin gene-related peptide as part of a protective network against hypertension, vascular hypertrophy, and oxidative stress. Hypertension. 2014;63:1056–1062.

148. Struthers AD, Brown MJ, MacDonald DW, et al. Human calcitonin gene related peptide: a potent endogenous vasodilator in man. Clin Sci. 1986;70:389–393.
149. Petersen KA, Birk S, Lassen LH, et al. The CGRP-antagonist, BIBN4096BS does not affect cerebral or systemic haemodynamics in healthy volunteers. Cephalalgia. 2005;25:139–147.
150. Berger A, Bloudek LM, Varon SF, et al. Adherence with migraine prophylaxis in clinical practice. Pain Practice. 2012;12:541–549.
151. Deen M, Edvige C, Katharina K, et al. Blocking CGRP in migraine patients–a review of pros and cons. J Head Pain. 2017;18:96.
152. Sarrouilhe D, Dejean C, Mesnil M. Involvement of gap junction channels in the pathophysiology of migraine with aura. Front Physiol. 2014;5:78.
153. Chan WN, Evans JM, Hadley MS, et al. Identification of (–)-cis-6-acetyl-4S-(3-chloro-4-fluoro-benzoylamino)-3, 4-dihydro-2, 2-dimethyl-2H-benzo [b] pyran-3S-ol as a potential antimigraine agent. Bioorg Med Chem Lett. 1999;9:285–290.
154. Smith MI, Read SJ, Chan WN, et al. Repetitive cortical spreading depression in a gyrencephalic feline brain: inhibition by the novel benzoylamino-benzopyran SB-220453. Cephalalgia. 2000;20:546–553.
155. Parsons AA, Bingham S, Raval P, et al. Tonabersat (SB-220453) a novel benzopyran with anticonvulsant properties attenuates trigeminal nerve-induced neurovascular reflexes. Br J Pharmacol. 2001;132:1549–1557.
156. Dahlof CGH, Hauge AW, Olesen J. Efficacy and safety of tonabersat, a gap-junction modulator, in the acute treatment of migraine: a double-blind, parallel-group, randomized study. Cephalalgia. 2009;29(Suppl 2):7–16.
157. Wang S-J, Chen P-K, Fuh J-L. Comorbidities of migraine. Front Neurol. 2010;1:16.
158. Napoletano F, Lionetto L, Martelletti P. Sumatriptan in clinical practice: effectiveness in migraine and the problem of psychiatric comorbidity. Expert Opin Pharmacother. 2014;15:303–305.
159. Lionetto L, Borro M, Curto M, et al. Choosing the safest acute therapy during chronic migraine prophylactic treatment: pharmacokinetic and pharmacodynamic considerations. Expert Opin Drug Metab Toxicol. 2016;12:399–406.
160. Benemei S, Cortese F, Labastida-Ramírez A, et al. School of Advanced Studies of the European Headache Federation (EHF-SAS). Triptans and CGRP blockade - impact on the cranial vasculature. J Headache Pain. 2017;18:103.
161. Capi M, Curto M, Lionetto L, et al. Eletriptan in the management of acute migraine: an update on the evidence for efficacy, safety, and consistent response. Ther Adv Neurol Disord. 2016;9:414–423.
162. Capi M, De Andrés F, Lionetto L,G, et al. Lasmiditan for the treatment of migraine. Expert Opin Investig Drugs. 2017;26:227–234.